

Dry Run Exercises: Solutions

Unit 6 While loops

Complete the dry run table for each fragment of code given in the questions below. **Give** the value(s) printed out when each is executed.

Follow the module dry run style guide in your answers.

1.

```
int count = 0;                                // L1

while(count < 3)                               // L2
{
    count = count + 1;                         // L3
    System.out.println(count);                 // L4
}
```

Complete the dry run table (you may not need all rows):

LINE	count	count < 3 ?	printed
L1	0		
L2		0 < 3 ? true	
L3	1		
L4			1
L2		1 < 3 ? true	
L3	2		
L4			2
L2		2 < 3 ? true	
L3	3		
L4			3
L2		3 < 3 ? false	

Write EXACTLY what is printed here

1
2
3

2.

```
String s = "ABA";           // L1

while(s.equals("ABA"))      // L2
{
    s = s + s;              // L3
}

System.out.println(s);      // L4
```

Complete the dry run table (you may not need all rows):

LINE	s	s.equals("ABA") ?	printed
L1	"ABA"		
L2		"ABA".equals("ABA") ? true	
L3	"ABAABA"		
L2		"ABAABA".equals("ABA") ? false	
L4			ABAABA

Write EXACTLY what is printed here

ABAABA

3.

```
int t = 3;                // L1

while(t<=8)              // L2
{
    t = t + 2;           // L3
}

System.out.println(t);    // L4
```

Complete the dry run table (you may not need all rows):

LINE	t	t <= 8 ?	printed
L1	3		
L2		3 <= 8 ? true	
L3	5		
L2		5 <= 8 ? true	
L3	7		
L2		7 <= 8 ? true	
L3	9		
L2		9 <= 8 ? false	
L4			9

Write EXACTLY what is printed here

9

4.

```
int t = 3;                // L1

while(t<=8)               // L2
{
    t = t + 2;            // L3
    System.out.println(t); // L4
}

System.out.println(t);    // L5
```

Complete the dry run table (you may not need all rows):

LINE	t	t <= 8	printed
L1	3		
L2		3 <= 8 ? true	
L3	5		
L4			5
L2		5 <= 8 ? true	
L3	7		
L4			7
L2		7 <= 8 ? true	
L3	9		
L4			9
L2		9 <= 8 ? false	
L5			9

Write EXACTLY what is printed here

5
7
9
9

5.

```
int n = 0; // L1

while(n <= 3) // L2
{
    System.out.print(n); // L3
    n = n + 2; // L4
}
System.out.print(n); // L5
```

Complete the dry run table (you may not need all rows):

LINE	n	n <= 3 ?	printed
L1	0		
L2		0 <= 3 ? true	
L3			0
L4	2		
L2		2 <= 3 ? true	
L3			2
L4	4		
L2		4 <= 3 ? false	
L5			4

Write EXACTLY what is printed here

024

6. In the following dry run assume that the numbers 18, 22, 21 are input when asked and that inputInt inputs an integer. after printing the String given it as an argument, then moving to a new line.

```
int age = inputInt("First?");    // L1

while(age != 21)                // L2
{
    System.out.println(age);    // L3
    age = inputInt("Next?");    // L4
}
System.out.println(age + " so stop."); // L5
```

Complete the dry run table (you may not need all rows):

LINE	age	age != 21 ?	printed
L1	18		First?
L2		18 != 21 ? true	
L3			18
L4	22		Next?
L2		22 != 21 ? true	
L3			22
L4	21		Next?
L2		21 != 21 ? false	
L5			21 so stop.

Write EXACTLY what is printed (note this means BY THE PROGRAM so not what typed by the user) here

```
First?
18
Next?
22
Next?
21 so stop.
```

7. In the following dry run assume that the numbers 18, 22, 21 are input when asked and that inputInt inputs an integer. after printing the String given it as an argument, then moving to a new line.

```
int age = inputInt("First?");    // L1

while(age != 21)                // L2
{
    age = inputInt("Next?");    // L3
}
System.out.println(age + " so stop."); // L4
```

Complete the dry run table (you may not need all rows):

LINE	age	age != 21 ?	printed
L1	18		First?
L2		18 != 21 ? true	
L3	22		Next?
L2		22 != 21 ? true	
L3	21		Next?
L2		21 != 21 ? false	
L4			21 so stop.

Write EXACTLY what is printed (note this means BY THE PROGRAM so not what typed by the user) here

First?
Next?
Next?
21 so stop.

8. In the following dry run assume that the numbers 18, 22, STOP are input when asked and that inputString inputs a String. after printing the String given it as an argument, then moving to a new line.

```
final String QUIT = "STOP";           // L1
String text = inputString("First?");  // L2

while(!(text.equals("STOP")))         // L3
{
    int age = Integer.parseInt(text); // L4
    text = inputString("Next?");      // L5
}
System.out.println(text + " so stop."); // L6
```

Complete the dry run table (you may not need all rows):

LINE	QUIT	text	age	!(text.equals("STOP")) ?	printed
L1	"STOP"				
L2		"18"			First?
L3				!("18".equals("STOP")) ? true	
L4			18		
L5		"22"			Next?
L3				!("22".equals("STOP")) ? true	
L4			22		
L5		"STOP"			Next?
L3				!("STOP".equals("STOP")) ? false	
L6					STOP so stop.

Write **EXACTLY** what is printed (note this means **BY THE PROGRAM** so not what typed by the user) here

First?
Next?
Next?
STOP so stop.